

# YINUO HAN

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## RESEARCH INTERESTS

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- Exoplanetary systems, debris disks, planet formation, disk substructure, dynamical evolution
- High-angular-resolution imaging, high-contrast optical/infrared imaging, millimeter interferometry
- Dust formation, carbon feedback, Wolf-Rayet stars, colliding-wind binaries

## ACADEMIC APPOINTMENTS

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1. **Melza M. and Frank Theodore Barr Fellow, Caltech, USA** Since Oct 2024  
Postdoctoral fellowship, Planetary Sciences, Division of Geological and Planetary Sciences

## EDUCATION

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1. **PhD in Astronomy, University of Cambridge, UK** Oct 2020 - Jul 2024  
Institute of Astronomy · Trinity College · Gates Cambridge Scholar  
**Thesis:** *The Structure of Extrasolar Planetesimal Belts in Images*  
**Advisor:** Prof. Mark Wyatt
2. **BSc (Adv) (Hons), University of Sydney, Australia** Feb 2016 - Dec 2019  
Majors in Physics, Neuroscience · First Class Honours (Physics) and the University Medal  
**Thesis:** *A Dance with Dragons: The Enigmatic Wolf-Rayet Binary Apep*  
**Advisor:** Prof. Peter Tuthill
- Study Abroad, Physics, University of California, Berkeley, USA** Aug 2017 - Dec 2017

## ADVISING

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List of theses and research projects advised.

- *VLT observations of  $\beta$  Pic and its dust clump.* Since Jun 2025  
Summer research project/Caltech SURF by Polaris Hayes (Caltech).
- *ALMA imaging of WR112 and dust modeling with JWST/MIRI.* Since Jun 2025  
Summer research project/Caltech SURF by Donglin Wu (Yale University).
- *Joint ALMA and JWST/NIRCam modeling of the vertical structure of the AU Mic debris disk.* Since Jun 2025  
Summer research project/Caltech SURF by Junyi Zhao (University of Michigan).
- *Secular perturbation as the origin of the dust and gas clump of Beta Pictoris.* Oct 2022 – Jun 2023  
Cambridge Part III/Masters project by Faran Sarwar (co-advised with Mark Wyatt).
- *The structure of the Beta Pictoris debris disk inferred from multi-wavelength mid-infrared images.* Jul – Aug 2022  
Summer research project by Andrew Zhang.

## TEACHING

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1. **Topics in Astrophysics** (Distributions, Timescales, Tides, Planet Formation) 2023  
Institute of Astronomy, University of Cambridge  
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
2. **Structure and Evolution of Stars** 2022  
Institute of Astronomy, University of Cambridge  
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
3. **Oscillating Systems, Waves and Quantum Waves, Fields** 2021  
Trinity College, University of Cambridge  
Weekly supervisions/tutorial classes for Part IA Physics/first-year undergraduate physics.
4. **Physics 1B** (Electromagnetism, Fluids, Quantum Mechanics) 2019  
School of Physics, University of Sydney  
Weekly tutorial classes for first-year undergraduate physics.

## OBSERVING

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- *ALMA co-I. program*: Testing a new picture of debris disk gas by observing CI towards HD 121617 (ID 2025.1.00273.S, P.I. Gianni Cataldi) C.12, 2025
- *VLT/VISIR P.I. program*: Testing the giant impact scenario to explain the dust clump and cat's tail in Beta Pictoris (114.27CQ) P114, 2024
- *ALMA P.I. program*: First resolved ALMA imaging of carbon-rich Wolf-Rayet binary dust structures (2024.1.00803.S) C.11, 2024
- *ALMA co-I. program*: Resolving the extended structure associated with Apep's enigmatic wind (2024.1.01244.S, P.I. Joseph Callingham) C.10, 2024
- *JWST P.I. program*: What lies beyond the inner spiral of Apep? (GO 5842) C.3, 2024
- *JWST P.I. program*: What causes warm dust interior to planetesimal belts? (GO 5709) C.3, 2024
- *VLT/VISIR P.I. program*: Wind kinematics in the extreme colliding-wind binary Apep: the first Galactic long-duration gamma-ray burst progenitor? (113.26A7) P113, 2024
- *ALMA P.I. program*: First ALMA imaging of dust around Wolf-Rayet binaries (2023.1.00999.S) C.10, 2023
- *ALMA co-I. program*: Vertical structure and planetary system dynamics (2023.1.00487.S and 2025.1.00362.S, P.I. Meredith Hughes) C.10/12, 2023/2025
- *JWST co-I. program*: Fingerprinting the history of episodic dust creation in Wolf-Rayet binaries (4093, P.I. Noel Richardson) C.2, 2023
- *Australia Telescope Compact Array*: Executing observations of Wolf-Rayet binary Apep 2019

## PRESS

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- *NRAO press release based on Donglin Wu's summer research published in Wu et al. 2026.* Feb 2026  
Yale press release.
- *Caltech press release based on Han et al. 2026.* Jan 2026  
Other ARKS collaboration press releases from NRAO, ALMA and ERC.
- *NASA, STScI and Caltech press releases based on Han et al. 2025.* Nov 2025  
Featured on Astronomy Picture of the Day.
- *University of Cambridge press release based on Han et al. 2022.* Oct 2022  
Reported by major science and mainstream news outlets (e.g., ABC, BBC, CNN, The Guardian).
- *University of Sydney press release based on Han et al. 2020.* Oct 2020  
Reported by 110 news items from 21 countries in 8 languages (e.g., CNN).

## ACADEMIC AWARDS

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### UNIVERSITY OF CAMBRIDGE

- Gates Cambridge Scholarship* 2020
- Cambridge International Scholarship*, accepted Gates above instead 2020

### UNIVERSITY OF SYDNEY

- University Medal*, top award among graduating class in physics 2019
- Henry Chamberlain Russell Prize for Astronomy*, for best astronomy Honours thesis 2019
- University of Sydney Honours Scholarship* 2019
- Faculty of Science Olympiad Scholarship* 2016 - 2019
- Deas-Thomson Scholarship in Physics*, for top of class in third-year physics 2018
- W.I.B. Smith Prize in Experimental Physics* 2018
- Dean's List of Academic Excellence* 2016, 2017, 2018
- Sydney Scholars Award* 2016 - 2018
- Denison Summer Research Scholarship* 2016, 2017
- Sydney Abroad International Exchange Scholarship*, supported second-year physics exchange 2017
- University of Sydney Academic Merit Prize* 2016
- Levey Scholarship in Physics*, for top of class in first-year physics 2016
- Smith Prize in Experimental Physics* 2016

### OTHER

- University of Oxford Clarendon Scholarship*, declined 2020
- International Physics Olympiad Bronze Medal* 2015

## RESEARCH GRANTS

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### EXTERNAL GRANTS

- P.I., JWST Program 5842 (USD 78k) 2024 – 2027
- P.I., JWST Program 5709 (USD 74k) 2024 – 2027

### INTERNAL GRANTS

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|----------------------------------------------------------------------------------------|-------------|
| Barr Fellowship research grant, Caltech (USD 30k)                                      | 2024 – 2027 |
| Gates Cambridge Scholarship research funding, University of Cambridge (approx. USD 3k) | 2020 – 2024 |
| Rouse Ball research funding, Trinity College, Cambridge (approx. USD 4k)               | 2023        |

## COLLABORATIONS

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List of large collaboration involvement working on major observational campaigns.

*ARKS*: ALMA Large Program to image debris disks at high resolution, P.I. Sebastian Marino

- *Role: Leading Radial Structure team. Our findings are available on [arkslp.org](http://arkslp.org).*

*JWST High Contrast Imaging*: JWST ERS program imaging exoplanets and disks, P.I. Sasha Hinkley

- *Role: Modeling MIRI 4-quadrant phase mask observations and VLT observations of HD 141569*

*DustERS*: JWST ERS program on Wolf-Rayet binary dust formation, P.I. Ryan Lau

- *Role: Developing spiral dust model for MIRI imaging published in Lau et al. 2022.*

## COMMUNITY INVOLVEMENT

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### REFEREE & PEER REVIEW

*Referee for MNRAS and AAS Journals*

*ALMA distributed peer review, Cycles 8 – 12*

*ESO distributed peer review*

### COMMUNITY & OUTREACH

*Caltech Summer Undergraduate Research Fellowship Reviewer* 2026

*Conference chair, Cambridge Interdisciplinary Forum for Postgraduate Research, University of Cambridge* 2024

*Public talk speaker, Institute of Astronomy, Cambridge open evenings* 2022 - 2023

*Technology Officer, Gates Cambridge Scholars Council* 2020 - 2021

*General Executive, Sydney University Mathematics Society* 2017 - 2019

*Ambassador (co-organizing outreach events), Australian Science Innovations (nonprofit in education)* 2016 - 2019

*Interviewer, Berkeley Scientific Journal, UC Berkeley* 2017

### OTHER EMPLOYMENT

*Supervisor (tutorial teaching), Institute of Astronomy, Cambridge* 2022 - 2023

*Supervisor (tutorial teaching), Trinity College, Cambridge* 2020 - 2021

*Tutor (tutorial teaching), School of Physics, University of Sydney* 2019

*Strategist, Goldman Sachs* 2018

## TALKS

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### CONFERENCE TALKS

- Exoplanets and Planet Formation Dec 2025  
*The high-resolution structure of debris disks in the ARKS ALMA program.* Shanghai, China

- Exploring Star–Planet–Disk Connections Oct 2025  
*High-resolution ALMA imaging of extrasolar planetesimal belts.* Tartu, Estonia
- Europlanet Science Congress 2024 Sep 2024  
*The high-resolution structure of debris disks revealed by the ARKS ALMA program.* Berlin, Germany
- Raising the veil on star formation near and far Apr 2024  
*High-resolution ALMA imaging of debris disks with the ARKS program.* Cambridge, UK
- Dust Devils: Debris Disks in the Sonoran Desert Mar 2024  
*The radial structure of debris disks in the ARKS ALMA program.* Tucson, USA
- Cosmic Dust 2023 Aug 2023  
*The physics of dusty Wolf-Rayet binaries revealed by observations of WR140.* Kitakyushu, Japan
- UK Exoplanet Community Meeting 2022 Sep 2022  
*Recovering the structure of edge-on debris disks non-parametrically.* Edinburgh, UK
- Debris Disks at Home and Abroad Sep 2022  
*Recovering the structure of edge-on debris disks non-parametrically.* Jena, Germany

## SEMINARS

- MPIA Planet and Star Formation Seminar Oct 2025  
*The formation and evolution of dust in the colliding-wind binary Apep revealed by JWST.* Heidelberg, Germany
- Tsinghua Exoplanets and Planet Formation Seminar Sep 2025  
*Imaging debris disks with ALMA and JWST.* Beijing, China
- Columbia Astronomy & Astrophysics Seminar Mar 2025  
*Resolving radial substructures in debris disks.* New York City, USA
- JPL Astrophysics Luncheon Seminar Feb 2025  
*Interpreting the Resolved Structure of Debris Disks.* Pasadena, USA
- Department of Applied Mathematics and Theoretical Physics, University of Cambridge Oct 2023  
*The radial and vertical structure of debris disks.* Cambridge, UK
- Institute of Astronomy, University of Cambridge Oct 2022  
*Stellar fireworks: the spiral dust plumes of Wolf-Rayet binaries.* Cambridge, UK
- Institute of Astronomy, University of Cambridge Mar 2022  
*A non-parametric method to recover the structure of edge-on debris disks.* Cambridge, UK

## OUTREACH TALKS

- Institute of Astronomy Open Evening Oct 2023  
*What are exoplanets doing? Asteroid belts give us a clue.* Cambridge, UK
- Institute of Astronomy Open Evening Jan 2023  
*Stellar fireworks: a journey through the spirally dust of massive binary stars.* Cambridge, UK
- Integrated Science Class Seminar (Year 6–8), Ming Cheng School Apr 2022  
*Planets of the Solar System and beyond.* Beijing, China
- Gates Cambridge Teach-a-thon Feb 2021  
*Planets of the Solar System and beyond.* Online

- Australian Council of Undergraduate Research  
*Highly anisotropic stellar winds in recently-identified Wolf-Rayet binary.* Oct 2019  
Newcastle, Australia

## POSTER PRESENTATIONS

- Gordon Research Conference on the Origins of Solar Systems  
*The radial structure of debris disks in the ARKS ALMA program.* Jun 2025  
South Hadley, USA
- Gordon Research Seminar + Conference on the Origins of Solar Systems  
*Modeling the structure of debris disks and the dust clump of Beta Pictoris.* Jun 2023  
South Hadley, USA
- STScI Spring Symposium: Planetary Systems and the Origins of Life in the Era of JWST  
*Modeling the structure of debris disks and the dust clump of Beta Pictoris.* May 2023  
Baltimore, USA
- Protostars and Planets VII  
*Has the dust clump in the debris disk of Beta Pictoris moved?* Apr 2023  
Kyoto, Japan

## PUBLICATIONS

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Publications available on ORCID 0000-0002-2106-0403, ADS Library and Google Scholar.

### FIRST-AUTHOR PUBLICATIONS

9. [In review] **Y. Han**, M. C. Wyatt, K. Y. L. Su, *et al.* A radially broad collisional cascade in the debris disk of  $\gamma$  Oph revealed by JWST. *Submitted to ApJ* (2026). doi.org/10.48550/arXiv.2601.15285
8. [In review] **Y. Han**, M. C. Wyatt, M. Jankovic, *et al.* The multiwavelength vertical structure of the archetypal  $\beta$  Pictoris debris disk. *Submitted to A&A* (2026). doi.org/10.48550/arXiv.2603.03540
7. **Y. Han**, E. Mansell, J. Jennings, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) II. The Radial Structure of Debris Disks. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556450
6. **Y. Han**, R. M. T. White, J. R. Callingham, *et al.* The formation and evolution of dust in the colliding-wind Wolf-Rayet binary Apep observed by JWST. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/ae12e5
5. **Y. Han**, M. C. Wyatt, S. Marino. Recovering the structure of debris disks non-parametrically from images. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf282
4. **Y. Han**, M. C. Wyatt, W. R. F. Dent. Has the dust clump in the debris disk of Beta Pictoris moved? *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stac3769
3. **Y. Han**, P. G. Tuthill, R. M. Lau, A. Soulain. Radiation-driven acceleration in the expanding WR140 dust shell. *Nature* (2022). doi.org/10.1038/s41586-022-05155-5
2. **Y. Han**, M. C. Wyatt, L. Matr . RAVE: a non-parametric method for recovering the surface brightness and height profiles of edge-on debris disks. *Monthly Notices of the Royal Astronomical Society* (2022). doi.org/10.1093/mnras/stac373

1. **Y. Han**, P. G. Tuthill, A. Soulain, J. R. Callingham, P. M. Williams, P. A. Crowther, B. J. S. Pope, B. Marcote. The extreme colliding-wind system Apep: resolved imagery of the central binary and dust plume in the infrared. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa2349

#### CO-AUTHORED PUBLICATIONS

\* Indicates projects led by students who I advised

24. \* D. Wu, **Y. Han**, *et al.* Constraining properties of dust formed in Wolf-Rayet binary WR 112 using mid-infrared and millimeter observations. *The Astrophysical Journal* (2026). doi.org/10.3847/1538-4357/ae31f1
23. S. Marino, L. Matrà, A. M. Hughes, J. Ehrhardt, G. M. Kennedy, C. del Burgo, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) I. Motivation, Sample, Data Reduction and Results Overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556489
22. B. Zawadzki, A. Fehr, A. M. Hughes, E. Mansell, J. Kittling, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) III. The Vertical Structure of Debris Discs. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556505
21. S. Mac Manamon, L. Matrà, S. Marino, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IV. CO gas imaging and overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556605
20. J. Milli, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) V. Comparison between scattered light and thermal emission. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556523
19. J. B. Lovell, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VI. Asymmetries and Offsets. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556568
18. A. Brennan, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VII. Optically Thick Gas with Broad CO Gaussian Local Line Profiles in the HD 121617 Disc. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556566
17. S. Marino, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VIII. A dust arc and non-Keplerian gas kinematics in HD 121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556493
16. P. Weber, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IX. Gas-driven origin for the continuum arc in the debris disc of HD121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556855
15. M. Jankovic, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) X. Interpreting the peculiar dust rings around HD 131835. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556637

14. R. M. T. White, B. J. S. Pope, P. G. Tuthill, **Y. Han**, *et al.* The Serpent Eating Its Own Tail: Dust Destruction in the Apep Colliding-Wind Nebula. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/adfbel
13. J. D. Monnier, **Y. Han**, *et al.* Revealing the Accelerating Wind in the Inner Region of Colliding-wind Binary WR 112. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-3881/adfa03
12. N. D. Richardson, *et al.* Carbon-rich dust injected into the interstellar medium by Galactic WC binaries survives for hundreds of years. *The Astrophysical Journal* (2025). 10.3847/1538-4357/addf30
11. M. Sommer, M. C. Wyatt, **Y. Han**. A PR drag origin for the Fomalhaut disk’s pervasive inner dust: constraints on collisional strengths, icy composition, and embedded planets. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf494
10. E. P. Lieb, *et al.* Dynamic imprints of colliding-wind dust formation from WR140. *The Astrophysical Journal Letters* (2025). doi.org/10.3847/2041-8213/ad9aa9
9. R. M. Lau, *et al.* A first look with JWST aperture masking interferometry (AMI): resolving circumstellar dust around the Wolf-Rayet binary WR 137 beyond the Rayleigh limit. *The Astrophysical Journal* (2024). doi.org/10.3847/2041-8213/ad9aa9
8. J. Terrill, S. Marino, R. A. Booth, **Y. Han**, J. Jennings, M. C. Wyatt. Deprojecting and constraining the vertical thickness of exoKuiper belts. *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stad1847
7. R. M. Lau, *et al.* From dust to nanodust: resolving circumstellar dust from the colliding-wind binary Wolf-Rayet 140. *The Astrophysical Journal* (2022). doi.org/10.3847/1538-4357/acd4c5
6. R. M. Lau, M. J. Hankins, **Y. Han**, *et al.* Nested dust shells around the Wolf-Rayet binary WR 140 observed with JWST. *Nature Astronomy* (2022). doi.org/10.1038/s41550-022-01812-x
5. P. A. Robinson, X. Gao, **Y. Han**. Relationships between lognormal distributions of neural properties, activity, criticality, and connectivity. *Biological Cybernetics* (2021). doi.org/10.1007/s00422-021-00871-z
4. B. Marcote, J. R. Callingham, M. De Becker, P. G. Edwards, **Y. Han**, R. Schulz, J. Stevens, P. G. Tuthill. AU-scale radio imaging of the wind collision region in the brightest and most luminous non-thermal colliding wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2021). doi.org/10.1093/mnras/staa3863
3. R. M. Lau, M. J. Hankins, **Y. Han** *et al.* Resolving periodic spirals and shadows from the Wolf-Rayet dust factory WR112. *The Astrophysical Journal* (2020). doi.org/10.3847/1538-4357/abaab8
2. J. R. Callingham, P. A. Crowther, P. M. Williams, P. G. Tuthill, **Y. Han**, B. J. S. Pope. Two Wolf-Rayet stars at the heart of colliding-wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa1244



1. M. Li, **Y. Han**, M. J. Aburn, M. Breakspear, R. Q. Poldrack, J. M. Shine, J. T. Lizier. Transitions in information processing dynamics at the whole-brain network level are driven by alterations in neural gain. *PLOS Computational Biology* (2019). doi.org/10.1371/journal.pcbi.1006957